

Chapter 10 Assignment — Systems of Equations and Inequalities

Name: _____

Date: _____

How to use this set: Each topic begins with *Strategy Notes* and a *Worked Example*. Then try the *Practice* (skill) and the *Challenge* (deeper) items. Give exact answers when possible. For matrix methods, show row operations; for graphing, label intercepts and vertices.

1. Topic 1: Systems in Two Variables

Strategy Notes. A system is linear if each equation is a linear combination of variables to the first power. Solve linear pairs by substitution or elimination; identify parallel/identical lines for none/infinitely many solutions.

Worked Example. Solve $\{x + 2y = 7, 3x - y = 1\}$. Multiply the second by 2 and add to the first: $(x + 2y) + (6x - 2y) = 7 + 2 \Rightarrow 7x = 9 \Rightarrow x = \frac{9}{7}$. Then $y = \frac{7 - \frac{9}{7}}{2} = \frac{\frac{20}{7}}{2} = \frac{10}{7}$.

Practice.

(a) Determine if the system is linear and solve if possible: $\{2x - y = 3, x^2 + y = 5\}$.

(b) Solve by elimination: $\{4x + 3y = 11, 2x - 5y = -13\}$.

Challenge.

- (a) For which real values of k does the system $\{x + ky = 4, 2x + 5y = 9\}$ have a unique solution? For which values does it have infinitely many or no solutions? When unique, solve in terms of k .

- (b) Without solving, decide whether the lines $3x - 2y = 6$ and $ax - \frac{2a}{3}y = 2a$ intersect for $a \neq 0$. Justify with a determinant.

2. Topic 2: Gaussian Elimination

Strategy Notes. Form the augmented matrix and use row operations to reach row-echelon form, then back-substitute. Be alert for inconsistent rows $[0 \ 0 \ \dots \mid k]$ and dependent systems with free variables.

Worked Example. For $\{x + y + z = 4, 2x - y + z = 1, x + 2y - 3z = 0\}$, reduce

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & -1 & 1 & 1 \\ 1 & 2 & -3 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -3 & -1 & -7 \\ 0 & 1 & -4 & -4 \end{array} \right] \rightarrow \dots$$

Continuing the elimination gives $(x, y, z) = \left(\frac{9}{13}, \frac{24}{13}, \frac{19}{13}\right)$.

Practice.

(a) Solve using Gaussian elimination: $\{x + 2y + z = 6, 2x - y + 3z = 5, 3x + y + 2z = 10\}$.

(b) Classify the solution set (unique/none/ininitely many) for $\{x + 2y - z = 1, 2x + 4y - 2z = 2, 3x + 6y - 3z = 3\}$.

Challenge.

- (a) Let $a \in \mathbb{R}$. Solve or classify the system $\{x + ay + z = 2, 2x + (2a - 1)y + 3z = 3, 3x + 3ay + (a + 1)z = 4\}$ in terms of a .
- (b) A cafe sells small, medium, and large smoothies that contain 1, 2, 3 scoops of fruit respectively and 1, 1, 2 cups of yogurt respectively. Total sales for a day were 40 small, 25 medium, and 15 large, bringing in \$505. If fruit costs \$0.80 per scoop and cups cost \$0.10 each, set up and solve a linear system to find the base price (without ingredients) and the profit per drink size.

3. Topic 3: Matrices and Row-Echelon Forms

Strategy Notes. In REF, all nonzero rows are above zero rows and leading entries move rightward; in RREF, each leading entry is 1 and is the only nonzero in its column.

Worked Example. $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$ is RREF. The solution of $[A|b]$ reads directly from the pivot columns; free variables correspond to non-pivot columns.

Practice.

(a) Reduce $\begin{bmatrix} 1 & 2 & -1 & 0 \\ 2 & 4 & 1 & 3 \\ -1 & -2 & 2 & -1 \end{bmatrix}$ to RREF.

(b) Decide whether $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix}$ is REF, RREF, or neither, and explain.

(a) Show that the rows of a square matrix are linearly dependent if and only if the columns are linearly dependent.

- (b) Find a matrix F such that for any 2×2 matrix A , the product AF swaps the columns of A . What is FA for your F ?

4. Topic 4: Matrix Algebra

Strategy Notes. Products are defined when inner dimensions match; $(AB)C = A(BC)$ but in general $AB \neq BA$. Transpose satisfies $(AB)^T = B^T A^T$.

Worked Example. With $A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \\ 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & 4 \\ 1 & 3 & 0 \end{bmatrix}$, $AB = \begin{bmatrix} 4 & 5 & 4 \\ -1 & -3 & 0 \\ 6 & -3 & 12 \end{bmatrix}$.

Practice.

(a) Compute AB for $A = \begin{bmatrix} 2 & -1 \\ 1 & 4 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 2 \\ -2 & 1 & 3 \end{bmatrix}$. Is BA defined?

(b) Verify $(AB)^T = B^T A^T$ for the matrices in part (a).

Challenge.

- (a) Let $M = \begin{bmatrix} 2 & 3 \\ -7 & 0 \end{bmatrix}$ and $N = \begin{bmatrix} -3 & -1 \\ 5 & 2 \end{bmatrix}$. Compute MN and NM and comment on whether $MN = NM$.

- (b) Given $R = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$ and $S = \begin{bmatrix} -1 & 5 \\ -6 & 3 \end{bmatrix}$, compute RS , $(RS)^T$, $R^T S^T$, and $S^T R^T$.
What do you notice?

5. **Topic 5: Inverses**

Strategy Notes. A 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible iff $ad - bc \neq 0$, with inverse $\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Solve $A\mathbf{x} = \mathbf{b}$ by $\mathbf{x} = A^{-1}\mathbf{b}$ when A is invertible.

Worked Example. For $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, $\det A = 5$, so $A^{-1} = \frac{1}{5} \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$. Then $A\mathbf{x} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$ gives $\mathbf{x} = \frac{1}{5} \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 7 \\ 5 \end{bmatrix} = \begin{bmatrix} \frac{13}{5} \\ \frac{3}{5} \end{bmatrix}$.

Practice.

- (a) Determine which of $\begin{bmatrix} 4 & 8 \\ 1 & 2 \end{bmatrix}$ and $\begin{bmatrix} 3 & 5 \\ 2 & 7 \end{bmatrix}$ is invertible, then find the inverse of the invertible one.

- (b) Solve $\{2x + y = 3, x - 3y = 5\}$ using a matrix inverse.

Challenge.

- (a) Exactly one of the matrices $A = \begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix}$, $B = \begin{bmatrix} 6 & -1 \\ 3 & 2 \end{bmatrix}$, $C = \begin{bmatrix} -2 & 3 \\ 1 & -1 \end{bmatrix}$ is invertible. Identify it using determinants and compute the inverse.

- (b) Solve $A\mathbf{x} = \mathbf{b}$ for $\mathbf{x}$ when $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 0 & -1 \\ -1 & 4 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 5 \\ -2 \end{bmatrix}$ using the inverse or elimination.

6. Topic 6: Determinants and Cramer's Rule

Strategy Notes. For 3×3 , expand by a convenient row/column or use row operations (tracking sign/scale). Cramer's Rule applies only when $\det A \neq 0$.

Worked Example. For $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 3 \\ 2 & 1 & 1 \end{bmatrix}$, expanding along the first row gives $\det A = 1 \cdot (-1 - 3) - 2 \cdot (0 - 6) = (-4) - (-12) = 8$.

Practice.

(a) Compute $\det \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \\ 5 & 2 & -2 \end{bmatrix}$.

(b) Use Cramer's Rule to solve $\{x + 2y + z = 4, 2x - y + z = 1, 3x + y - 2z = 0\}$.

Challenge.

(a) Evaluate $\begin{vmatrix} 0.27 & -175 \\ -0.15 & 95 \end{vmatrix}$ exactly.

(b) For each matrix $\begin{bmatrix} -2 & 3 \\ -2 & 8 \end{bmatrix}$, $\begin{bmatrix} 0 & -3 \\ -2 & 0 \end{bmatrix}$, and $\begin{bmatrix} 6 & 12 \\ -2 & -4 \end{bmatrix}$, determine whether an inverse exists and, if so, find it.

7. Topic 7: Partial Fraction Decomposition

Strategy Notes. For distinct linear factors, write $\frac{P(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$ and equate coefficients or use cover-up.

Worked Example. $\frac{5x+3}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$. Setting $x = 1$ gives $A = \frac{8}{3}$; $x = -2$ gives $B = \frac{7}{3}$.

Practice.

(a) Find A, B so that $\frac{7x-4}{(x+1)(x-3)} = \frac{A}{x+1} + \frac{B}{x-3}$.

(b) Decompose $\frac{3x+2}{(x-2)^2}$ into the form $\frac{A}{x-2} + \frac{B}{(x-2)^2}$.

8. Topic 8: Nonlinear Systems

Strategy Notes. Combine substitution with identities (e.g., completing the square) or use resultants/symmetry for xy terms.

Worked Example. Solve $\{x + y = 6, xy = 8\}$. Then $t = x$ and $6 - t = y$, so $t(6 - t) = 8 \Rightarrow t^2 - 6t + 8 = 0 \Rightarrow t = 2, 4$, hence $(x, y) = (2, 4)$ or $(4, 2)$.

Practice.

(a) Solve $\{x^2 + y^2 = 13, y = 1 - x\}$.

(b) Solve $\{x + y = 1, xy = -6\}$.

9. Topic 9: Systems of Inequalities and Linear Programming

Strategy Notes. Graph half-planes, find intersection polygon, and evaluate the linear objective at vertices.

Worked Example. Maximize $P = 3x + 2y$ subject to $x + 2y \leq 8$, $2x + y \leq 10$, $x, y \geq 0$. Vertices are $(0, 0)$, $(0, 4)$, $(4, 2)$, $(5, 0)$; maximum $P = 16$ at $(x, y) = (4, 2)$.

Practice.

(a) Graph the solution of $\{x + y \leq 6, x - 2y \geq -4, x \geq 0, y \geq 0\}$ and list its vertices.

(b) A shop sells tables (profit \$40) and shelves (profit \$25). Each table uses 3 hours carpentry and 1 hour finishing; each shelf uses 2 hours carpentry and 1 hour finishing. With at most 24 carpentry hours and 10 finishing hours, find the production plan that maximizes profit.